



MEASUREMENT & UNITS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Fundamental Physical Quantities — SI Base Concept

- Physics mein 7 fundamental physical quantities hain — length, mass, time, electric current, temperature, luminous intensity aur amount of substance. Inhi ko base/fundamental units mein measure kiya jata hai. [Q1]
- Given options mein electric current fundamental quantity hai; force, velocity aur work derived quantities hain. [Q1]

2. Force, Work, Power & Energy — Core Units

- Power = work/time; SI unit watt (W) hai aur $1\text{ W} = 1\text{ J/s}$. Electric power ki unit bhi watt hi hoti hai. [Q2, Q3]
- Force motion ko change karne wali interaction hai; $F = ma$ aur SI unit newton (N) = $\text{kg}\cdot\text{m/s}^2$. [Q4]
- Work = force $\times$ displacement in force direction; work scalar quantity hai. SI unit joule (J) = $\text{N}\cdot\text{m}$ hai; joule energy ki bhi unit hai. [Q5]
- Power-plant context mein megawatt power ko show karta hai: $1\text{ MW} = 10^6\text{ W}$. Horsepower bhi power ki unit hai: $1\text{ hp} \approx 746\text{ W}$. [Q24, Q29]
- Standard matching: acceleration $\rightarrow \text{m/s}^2$, force $\rightarrow$ newton, work done $\rightarrow$ joule, impulse $\rightarrow$ newton-second ($\text{N}\cdot\text{s}$). [Q25]

3. Frequency, Hertz & GHz — Cycles and CPU Speed

- Frequency ki SI derived unit hertz (Hz) hai; $1\text{ Hz} = 1\text{ cycle per second}$. Isliye “what is measured in hertz?” ka answer frequency hai. [Q6, Q7]
- $1\text{ GHz} = 10^9\text{ Hz}$. Computer mein GHz mainly CPU clock/processing speed ko indicate karta hai — CPU per second kitne cycles execute karta hai. [Q8]

4. Electrical Units — Ampere, Volt, Watt & Resistivity

- Electrical resistivity material ka intrinsic property hai jo current flow ke opposition ko quantify karta hai; $\rho = RA/l$ aur SI unit ohm-metre ($\Omega\cdot\text{m}$) hai. [Q9, Q10]
- Ohm (Ω) resistance ki unit hai, jabki ohm-metre ($\Omega\cdot\text{m}$) resistivity ki unit hai — dono ko interchange nahi karna chahiye. [Q9, Q10]
- Ampere electric current ki SI unit hai. Source ke according 1 A current roughly 6.25×10^{18} electrons per second ke charge flow ke equivalent hai. [Q23]

- Electric-unit map yaad rakhein: joule → work, ampere → current, watt → power, volt → electric potential. [Q28]

⚠ **EXAM TRAP — Resistance vs Resistivity — Q9–Q10**

- ▶ Q9 disputed hai kyunki correct SI unit resistivity ke liye $\Omega \cdot m$ hai, lekin option list mein sirf ohm diya gaya tha; isliye source ne answer (*) mark kiya. [Q9]
- ▶ Exam memory: Resistance → Ω ; Resistivity → $\Omega \cdot m$. [Q9, Q10]

5. Magnetic Field — Tesla and Equivalent Units

- Magnetic flux density / magnetic B-field ki SI unit tesla (T) hai. Magnetic field strength H ki unit A/m hoti hai. [Q11]
- 1 tesla = 1 weber/m² = 1 newton/(ampere·metre). Unit Nikola Tesla ke naam par hai. [Q12]

6. Radioactivity — Becquerel vs Curie

- Radioactive activity ki SI unit becquerel (Bq) hai; 1 Bq ka meaning 1 decay per second hai. Iska naam Henri Becquerel ke naam par hai. [Q13]
- Curie (Ci) radioactivity ki old/non-SI unit hai; isi wajah se “Activity of radioactive substance — Curie” SI-unit matching mein incorrect pair hai. [Q13]

⚠ **EXAM TRAP — Becquerel vs Curie — Q13**

- ▶ SI radioactivity → Becquerel (Bq); Curie (Ci) non-SI/older unit hai. [Q13]

7. Astronomical Distance — Light-Year & Parsec

- Light-year time ki unit nahi, astronomical distance ki unit hai — light vacuum mein one year mein jitna distance travel karti hai. 1 ly $\approx 9.46 \times 10^{15}$ m. [Q14, Q15, Q16]
- Astronomical distances ko light-year mein express karne ka key reason light ki vacuum speed ka constant hona hai. [Q17]
- Parsec bhi astronomical distance ki unit hai: 1 pc $\approx 3.0857 \times 10^{16}$ m ≈ 3.26 light-years; source option mein ≈ 3.25 ly diya gaya. [Q18, Q19]
- “Light-year — unit of measuring time” incorrect matching hai; correct relation distance se hai. [Q30]

⚠ **EXAM TRAP — Light-Year ≠ Time — Q14–Q19, Q30**

- ▶ Naam mein “year” hone ke bawajood light-year DISTANCE ki unit hai. [Q14, Q15, Q16, Q30]
- ▶ Parsec bhi distance ki unit hai; 1 parsec ≈ 3.26 light-years. [Q18, Q19]

8. Length Conversions — nm, Micron, Angstrom, Fermi & km

- Centimetre to inch conversion ke liye source relation: 1 cm ≈ 0.39 inch; centimetre value ko 0.39 se multiply karke approximate inches milte hain. [Q20]
- 1 nm = 10^{-9} m = 10^{-7} cm. Source conversion ke hisab se 6 ft ≈ 1.83 m $\approx 183 \times 10^7$ nm. [Q21, Q22]
- Angstrom (Å) microscopic length/wavelength ke liye use hota hai: 1 Å = 10^{-10} m = 0.1 nm. [Q31]
- 1 micron (μm) = 10^{-6} m = 10^{-3} mm = 10^{-4} cm. [Q34, Q35]
- Basic conversion: 1 km = 1000 m. [Q39]
- Given units mein size order: micron 10^{-6} m > nanometre 10^{-9} m > angstrom 10^{-10} m > fermi/femtometre 10^{-15} m; smallest = fermi. [Q49]

9. Match-the-Following — Common Quantity–Unit Pairs

- Pressure–Dyne incorrect SI matching hai: pressure/stress ki SI unit pascal (Pa) hoti hai; dyne CGS system ki force unit hai. [Q26]
- Watt → power; knot → ship speed; nautical mile → sea/navigation distance; calorie → heat/energy. [Q27]
- High speed → Mach; wavelength → angstrom; pressure → pascal; energy → joule. [Q32]
- Analogy: Joule : Energy :: Pascal : Pressure. $1 \text{ Pa} = 1 \text{ N/m}^2$. [Q33]

10. Heat, Temperature & Power — Do Not Mix Them

- Celsius/Centigrade temperature ka scale/unit hai, heat ki unit nahi. Heat-energy ke common units calorie, kilocalorie, joule/kilojoule aur erg hain. [Q36, Q38]
- Watt heat ki unit nahi — watt power ki unit hai. Calorie, kilocalorie aur kilojoule heat/energy units hain. [Q37]
- Source Q36 context: nautical mile sea distance ke liye; 1 nautical mile $\approx 1.852 \text{ km}$. Horsepower power ki unit hai; $1 \text{ hp} \approx 746 \text{ W}$. [Q36]

⚠ EXAM TRAP — Heat vs Temperature vs Power — Q36–Q38

- ▶ Celsius/Centigrade → temperature; Watt → power; Calorie/Joule/Kilojoule/Erg → heat-energy. [Q36, Q37, Q38]

11. Pressure Units — Pascal, Bar, atm, Torr & kgf/cm²

- Pascal (Pa) pressure aur stress ki SI unit hai; $1 \text{ Pa} = 1 \text{ N/m}^2$. Pressure ka dimensional SI form $\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}$ hota hai. [Q41, Q42]
- bar, pascal (Pa), torr aur atmosphere (atm) — sab pressure units hain. $1 \text{ bar} = 10^5 \text{ Pa}$ aur $1 \text{ atm} = 101,325 \text{ Pa}$. [Q44, Q45]
- “kg/cm²” technically proper pressure notation nahi; intended form kgf/cm² hota hai. $1 \text{ kgf/cm}^2 \approx 98,066.5 \text{ Pa}$. [Q43]
- Approximate exam conversion: $1 \text{ kgf/cm}^2 \approx 1.0 \text{ bar}$; more precisely about 0.98 bar. [Q46]

⚠ EXAM TRAP — Pressure Notation & Units — Q43–Q46

- ▶ SI pressure → Pascal, not bar. Bar metric but non-SI unit hai. [Q41, Q42, Q45]
- ▶ kg/cm² ko pressure ke liye casually likha jata hai, lekin force notation kgf/cm² technically correct hai. [Q43, Q46]

12. Dimensionless Quantities — Strain & Mach

- Strain = change in dimension / original dimension. Ratio hone ki wajah se strain dimensionless hai aur iski koi unit nahi hoti. [Q47]
- Mach number bhi dimensionless ratio hai: object speed / local speed of sound; Mach 1 = speed of sound. [Q32]

13. Mass Prefix — Picogram

- Picogram (pg) bahut chhoti mass unit hai: $1 \text{ pg} = 10^{-12} \text{ gram}$. Comparison: $1 \text{ mg} = 10^{-3} \text{ g}$; microgram = 10^{-6} g ; femtogram = 10^{-15} g . [Q40]

14. Oil Measurement — Barrel Conversion

- Oil-volume convention: $1 \text{ barrel} \approx 158.9873 \text{ litre} \approx 159 \text{ litre} = 0.158987 \text{ m}^3 = 42 \text{ U.S. gallons}$. [Q48]

15. Special Units — Cusec, Byte, Richter, Bar & NTU

- Mixed-unit matching: Cusec → rate of flow; Byte → computer/digital information; Richter → earthquake magnitude/intensity; Bar → pressure. [Q50]

- Cusec = cubic foot per second; water-flow rate ke liye use hota hai. $1 \text{ cusec} \approx 28.317 \text{ litres/second}$. [Q50, Q51]
- NTU = Nephelometric Turbidity Unit; water ki turbidity/gandlapan ko measure karta hai. Suspended solids badhenge to turbidity value bhi badhegi. [Q52]

16. Solar Constant — Approximate Exam Value

- Solar constant/TSI roughly Sun se 1 astronomical unit distance par per unit area received solar power ko represent karta hai. [Q53]
- Source explanation modern measurement $\approx 1361 \text{ W/m}^2$ batati hai; given options mein closest exam answer 1.4 kW/m^2 ($\approx 1400 \text{ W/m}^2$) tha. [Q53]

EXAM TRAP — Solar Constant — Exact vs Exam Approximation — Q53

- ▶ Source question ke options mein 1.4 kW/m^2 closest answer hai; explanation $\sim 1361 \text{ W/m}^2$ ka measured value deti hai. [Q53]

17. Dobson Unit — Ozone Layer Thickness / Concentration

- Ozone layer thickness/concentration Dobson Unit (DU) mein express ki jati hai. $1 \text{ DU} = \text{STP par } 0.01 \text{ mm ozone thickness}$. [Q54, Q55, Q56, Q57]
- $1 \text{ DU} \approx 2.687 \times 10^{20}$ ozone molecules per square metre. Unit ka naam atmospheric-ozone scientist G.M.B. Dobson ke naam par hai. [Q54, Q57]
- Exam wording “ozone thickness”, “ozone layer thickness” aur “ozone concentration” — tino contexts mein Dobson unit ko identify karna hai. [Q54, Q55, Q56, Q57]